

# Geological setting, age and endowment of major Australian mineral deposits - a compilation (2024 update)

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D. L. Huston<sup>1</sup>, J. Cloutier<sup>1</sup>, A. Burnham<sup>1</sup>, Y. Cheng<sup>1</sup>, M. P. Doublier<sup>1</sup>, and P. M. Downes<sup>2</sup>

<sup>1</sup> Geoscience Australia

<sup>2</sup> Formerly Geological Survey of New South Wales

**Department of Industry, Science, Energy and Resources**

Minister for Resources, Water and Northern Australia: The Hon Madeleine King MP

Secretary: Mr David Fredericks PSM

**Geoscience Australia**

Chief Executive Officer: Dr James Johnson

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# Contents

|                                                             |    |
|-------------------------------------------------------------|----|
| 1 Summary .....                                             | 1  |
| 2 Introduction .....                                        | 2  |
| 3 Dataset description .....                                 | 4  |
| 3.1 Name of deposit/district and location information ..... | 4  |
| 3.2 Tectonic provinces .....                                | 4  |
| 3.3 Mineralising events and their age .....                 | 5  |
| 3.3.1 Age of mineralisation .....                           | 5  |
| 3.4 Endowment .....                                         | 7  |
| 3.5 Major and minor products .....                          | 8  |
| 3.6 Host rocks .....                                        | 8  |
| 3.7 Associated igneous rocks .....                          | 8  |
| 3.8 Alteration assemblages and zonation .....               | 9  |
| 3.9 Fe-S-O minerals .....                                   | 9  |
| 3.10 Sulfate minerals .....                                 | 9  |
| 3.11 Metamorphic grade .....                                | 9  |
| 3.12 Data sources .....                                     | 9  |
| 3.13 Comments .....                                         | 10 |
| 4 Acknowledgements .....                                    | 11 |
| 5 References .....                                          | 12 |



# 1 Summary

Herein we present the results of an updated national compilation of mineral deposits (available in Excel or CSV format) for Australia. The deposits were selected as they have substantial endowment (i.e., pre-mining mineral resource) and/or detailed geological information is available. For each deposit (or, in some cases, district) the dataset includes information on:

1. Name (including synonyms), location and GA identifying numbers;
2. Tectonic province that hosts the deposit;
3. Type(s) and age(s) of mineralising events that produced/affected the deposit (including metadata on ages);
4. The geological endowment of the deposit;
5. A summary of known and potential major (main and co-) and minor (by-) products based on economic value;
6. The host rocks to the deposit;
7. Spatially and/or temporally associated magmatic rocks;
8. Spatially and temporally associated alteration assemblages (mostly proximal, but, in some cases, regional assemblages);
9. The Fe-S-O minerals present in the deposit and their relative abundances;
10. Sulfate minerals present;
11. Metamorphic grade;
12. Data sources; and
13. Comments.

This document presents more detailed descriptions of the metadata presented in the compilation. The dataset is presented in Appendix A; Appendix B presents a national classification of geological provinces based mostly on existing State survey classifications; Appendix C presents a deposit classification based on the classification proposed by Hofstra et al. (2021); Appendix D presents mineral abbreviations used in the dataset; and Appendix E presents value calculations to determined major and minor commodities.

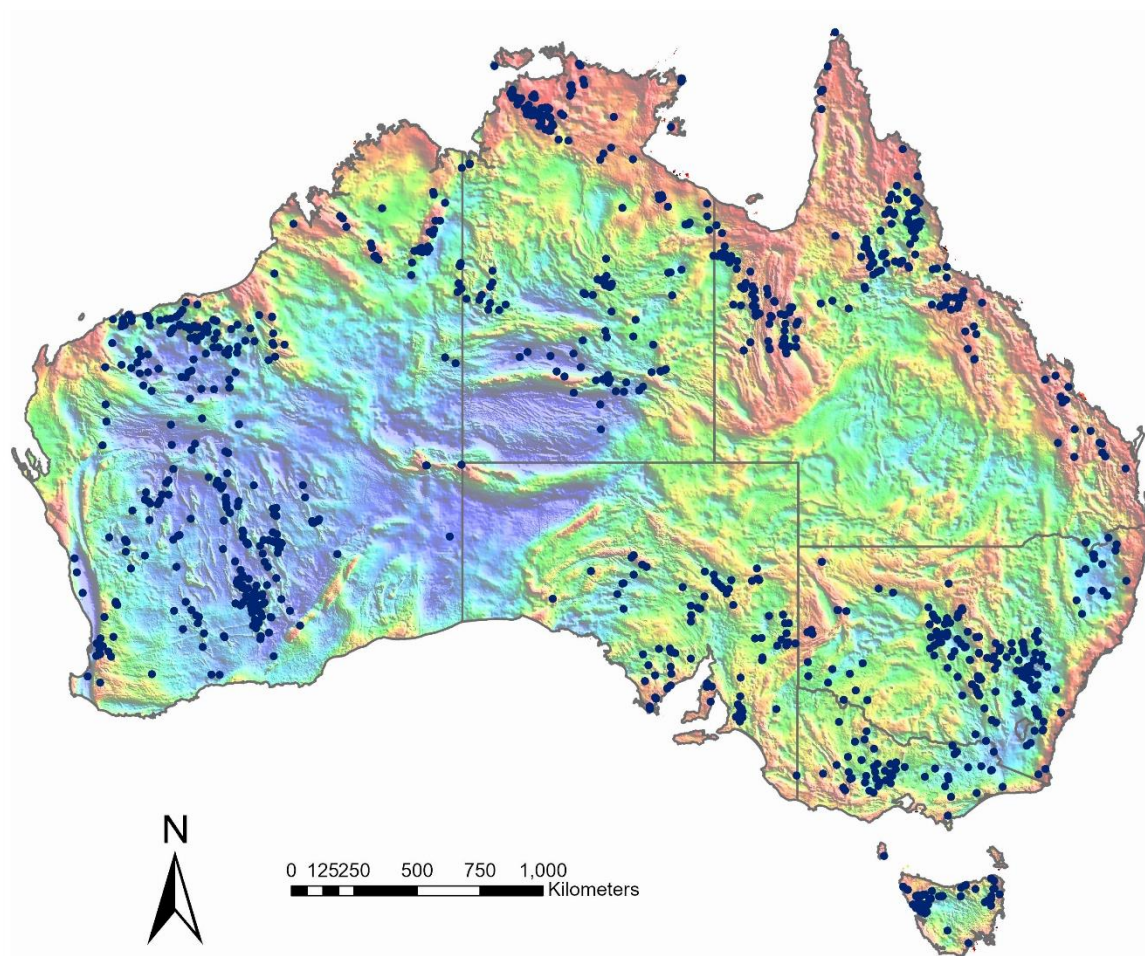
## 2 Introduction

Constraining the distribution of mineral deposits in space and in time is essential to understanding mineral systems. Although the distribution of deposits in space has been documented in Australia through national-scale datasets such as OZMIN (Ewers and Ryburn, 1994) and MINLOC, the fourth dimension, time, is less well documented. With the advent of a range of relatively robust and inexpensive geochronology tools, the understanding of the timing of mineralisation has improved markedly. This potentially enables analysis that not only includes spatial information, but also temporal information. Before this potential is fully realised and applied to ore genesis studies and exploration, spatial and temporal data, along with information such as deposit classification, metal endowment, host rock, associated magmatic rocks and other information, must be assembled.

The accompanying dataset (Appendix A) is updated from the first compilation produced in 2021 (Huston et al., 2021) and has been increased by about 10% to include new discoveries and other deposits for which new data have become available. The original dataset (Huston et al., 2021) has been updated to include recent data on the geology, geochronology and mineral resources, and to rectify errors in the original dataset. The dataset is a comprehensive, though not exhaustive, compilation of selected attributes of important mineral deposits and districts in Australia. The purpose is to capture data about significant mineral deposits. The term “significant” generally relates to the size of the mineral endowment (contained pre-mining resources). For many commodities there are many hundreds or thousands of deposits and occurrences around Australia, with only a small fraction of these deposits/occurrences making a significant contribution to Australia’s mineral endowment. The dataset contains information about these major deposits. In some cases, a number of small deposits have been grouped together into a district, but in other cases, small deposits have been ignored. However, where important information, such as age of small deposits or occurrences are available, they have been captured. Figure 1 shows the locations of deposits incorporated into the updated dataset.

The structure of the dataset has also been modified. These modifications include: (1) incorporation of additional commodities (individual rare earth elements and Rb); (2) indication if mineral resources reported are compliant with CRIRSCO (mostly JORC Code 2012) codes; (3) identification of major (main- and co-) and minor (both known and potential) commodities in the ores; and (4) separating references into those related to location, geology, geochronology and endowment. This last structural modification enables more precise access to references used in this dataset. Finally, rather than listing the references at the end of this document, the references are presented in a second tab in Appendix A.

In addition to these changes to the deposit dataset (Appendix A), the supporting documents (Appendices B-D) have also been updated to reflect updated province and mineral deposit classification schemes, and mineral abbreviations. A new supporting document (Appendix E) has been added to show calculations made to determine major and minor commodities.



*Figure 1. Location of mineral deposits and occurrences documented in this dataset overlain on a gravity image of Australia (Lane et al., 2020).*

## 3 Dataset description

The dataset provides (1) a name (and synonyms) of each deposit/occurrence, (2) location information, (3) information on the tectonic province that hosts the deposit, (4) information on mineralising event(s) that formed the deposit, (5) information on the age of these events (including an assessment of the reliability of the age), (6) information on mineral endowment (including grade and tonnage data), (7) identification of known and potential major (main and co-) and minor (by-) products, (8) information on the host(s) of the deposit and associated igneous rocks, (9) details of regional and deposit-scale alteration assemblages, including spatial and temporal zonation, (10) Fe-S-O mineralogy, (11) the presence and abundance of sulfate minerals, (12) metamorphic grade, (13) data sources, and (14) detailed comments about the data presented.

This dataset is intended to be “living” in that it will be updated as more information becomes available, new deposits are discovered, and errors are found and corrected. Users are asked to contact Geoscience Australia client services at [clientservices@ga.gov.au](mailto:clientservices@ga.gov.au) to correct errors or to suggest to add new information.

### 3.1 Name of deposit/district and location information

For each entry in the dataset a name is provided. In most cases this is the name of an individual deposit, but in some cases, it was not possible to separate individual deposits with respect to resource data. In these cases, the individual deposits have been combined into districts (or camps) and data reported for the district, and individual deposits included in the district are listed in parentheses. In most cases, Geoscience Australia OZMIN (ENO) or MINLOC (MINLOCNO) database entity numbers have been assigned to entries, and the associated location has been used. In instances where the entry is not in OZMIN or MINLOC, location information has been derived either from State/Territory mineral occurrence datasets, from scientific literature, company announcements to stock exchanges (e.g., Australian Securities Exchange or Toronto Stock Exchange), or from the location of the drill hole from which age information cited was extracted. In cases where the entity is a district, the approximate geographic centre or the most significant deposit was used as the location. Where not combined at the district-scale to calculate endowment, individual deposits have been grouped into geographical districts.

The source of the location is specifically identified in the “Source(s) of data – location” column. In cases where the source is [www.portal.ga.gov](http://www.portal.ga.gov), the location can be sourced in the “Deposit Summary” dropdown in Tools. The location and some other information can be extracted by a database search using the cited ENO.

### 3.2 Tectonic provinces

Each individual entry has been assigned to scaled tectonic entities. At the largest scale, five tectonic elements were used: (1) West Australian Element, (2) North Australian Element, (3) South Australian Element, (4) Central Australian Element, and (5) Tasman Element. These elements follow the scheme proposed by Blewett et al. (2012). The next scale down is termed a ‘province’, which can be either cratonic blocks (e.g., Yilgarn, Pilbara, Gawler, etc.), major provinces within crustal elements (e.g., Pine Creek, Aileron, Warumpi, etc.) major orogenic provinces (e.g., Capricorn, Delamarian, Lachlan, etc.),



or major basins or basin systems (e.g., Hamersley, North Australian Basin System, Centralian Basin System, Canning, etc.). Provinces are subdivided into subprovinces, which are then divided into zones. In cases where basins extend across element borders, the element immediately underlying the basin at the deposit location was used as the highest level in the tectonic hierarchy. This scheme is detailed in Appendix B and is intended to be compatible as much as possible with existing state schema.

### 3.3 Mineralising events and their age

Mineral deposits commonly form through a series of events that introduce metal and subsequently enrich metal concentrations or otherwise upgrade the mineralised rock to a point that allows feasible metal extraction (Huston et al., 2016). Because deposits commonly form as a series of events, the dataset provides up to three mineralising events for each entry. Each mineralising event includes the relative timing of the event, the relative contribution of each event to the final mineralised rock, classification of the event in a mineral system framework and information about the age of the event.

The relative timing of the event can be precursor, primary or secondary. A precursor timing indicates the formation of proto-ore that requires upgrading prior to economically viable extraction. An example of a precursor would be banded iron formation, which requires hypogene or supergene upgrading to become iron ore. A primary timing indicates the event that initially formed ore, and a secondary timing indicates an event that overprinted the ores. Although secondary events are commonly related to surficial weathering, they also include events such as redistribution associated with metamorphism. However, subsequent hydrothermal events inferred to introduce new metal are also considered primary.

The relative contribution of the event to ore formation is a qualitative assessment of the importance of the event to the deposit. Six categories can be assigned to this criterion: total, dominant, significant, minor, precursor and residual. A total contribution implies all ore was produced during the event. Dominant indicates that the event contributed at least 50% of metal to the final mineralised body of rocks or was the most important process upgrading the mineralised body. Significant indicates that the event contributed 10-50% of metal or had a secondary role in upgrading the mineralised body. Minor indicates that the event contributed less than 10% of the metal or had a minor role in upgrading. *Precursor* indicates the ore that was present during the precursor timing (above). *Residual* indicates that the event has been virtually or totally destroyed by subsequent events, with only vestigial evidence of previous existence.

Each event has been assigned a deposit type, which fits into a classification scheme that is tiered from deposit type to mineral environment to mineral system. This system is modified from a classification scheme developed by a consortium involving Geoscience Australia, the Geological Survey of Canada and the United States Geological Survey with additional contributions from the Geological Survey of Queensland (Hofstra et al., 2021). As much as possible the classification is based on observed characteristics of the deposits, including lithological associations, form and metal assemblage, among others. Appendix C summarises the classification scheme and presents common synonyms of the deposit types.

#### 3.3.1 Age of mineralisation

Most mineralising events have been assigned an age. In many cases these ages are based on radiogenic isotope data, but in other cases the ages are based on correlations with other similar deposits that have good age control or other geological constraints, including faunal assemblages with

host units, paleomagnetic correlations and geological relationships. Age information presented includes the assigned age and uncertainty (in Ma). For radiogenic age determinations the cited age commonly is a weighted mean age with uncertainty at the  $2\sigma$  level (~95%). In cases where geological relationships are used along with radiogenic and other ages, the ages assigned are midpoints of possible ranges, and uncertainties are assigned to include the entire age range. In some cases, such as model ages or correlation ages, uncertainty estimates are not provided as they cannot be reliably estimated.

In addition to the age and uncertainty, metadata on the age are provided, including type, basis, mineral dated, age system, radiometric age type, age instrumentation and age confidence. These metadata are provided to allow the user to assess the reliability of the assigned age. In a broad sense, there are two types of age determinations, direct and inferred. A direct age, which, most commonly, is an age determined directly from an ore mineral, requires no assumptions to assign an age as the age of mineralisation.

An inferred age requires an assumption about the relationship of the reported age and the mineralising event. In some cases, datable minerals are intergrown with the ore mineral assemblage but are not themselves ore minerals. In this case there is an assumption that the ore assemblage and datable mineral grew concurrently. An inference for some deposits is that the deposit is syngenetic or early diagenetic. In this case the age of the host unit or succession is the same as or closely approximates the age of mineralisation. Another common inference that is made is that a datable alteration assemblage formed at the same time as mineralisation. In most cases this is a robust assumption, but in others the relationship is not as clear. A third common inference is that a datable magmatic rock (e.g., a granite) is temporally and genetically linked to mineralisation. In addition to these three major inferences, there are a number of other inferences that are made in some other instances. An important inference for Mississippi Valley-type and siliciclastic-hosted base-metal deposits is that a geological process that produces a paleomagnetic pole can be related to mineralisation. The paleomagnetic dating method also assumes that the pole can be related to a well constrained local apparent polar wander path. In some cases, geological relationships have been used to assign ages. In these cases, details of these relationships are provided in comments.

In a small percentage of deposits, a reference may provide an age estimate without supporting data as to its origin. If no other age data are available for that deposit, this age estimate is reported without uncertainty and with an age type of uncertain.

For ages based on radiogenic isotope determinations, information on the mineral(s) dated and isotopic system used to determine the age is provided. In addition, information on the age type (e.g., pooled, isochron, model, etc.), and instrumentation (e.g., SHRIMP, TIMS, LA-ICP-MS, etc.) are provided.

For each cited age, the compilers have assessed its reliability. The reliability parameter ranges from 0 to 5, with a value of 5 having the greatest confidence. A value of 5 was only assigned for direct ages with robust isotopic methods. An example of an age with a reliability parameter of 5 is a Re-Os molybdenite age. A reliability parameter of 5 indicates the age is very highly likely to be correct.

Values of 4 were assigned to ages for which a robust inference can be made with regards to the relationship of mineralisation to the event dated and the method of measurement is robust. An example with a reliability parameter of 4 would be an age of a volcanic rock closely associated with a syngenetic deposit. A reliability parameter of 4 indicates the age is highly likely to be correct.

Values of 3 were assigned to ages for which the inference is not as strong or in which there is uncertainty in interpreting the isotopic or other data. Examples include a  $^{40}\text{Ar}/^{39}\text{Ar}$  age of white mica

alteration or a lead isotope model age from a well-constrained evolution model. A reliability parameter of 3 indicates the age is very likely to be correct.

Values of 2 were assigned to ages for which the strength of inference is moderate and/or the robustness of the analytical method is not that strong. An example of this would be that the deposit of interest has a similar age to a nearby deposit with a well-constrained age. Another example would be an age based upon the relationship of a paleomagnetic pole to a well-constrained paleomagnetic wander path. A reliability parameter of 2 indicates the age is likely to be correct.

Values of 1 were assigned to ages for which there is significant uncertainty as to the inference and/or the robustness of the analytical method. An example of this would be a reported age that lacks sufficient details upon which a reliability can be assessed. In many cases these are ages reported in abstracts. A reliability parameter of 1 indicates the age may be correct.

In very few cases, reliability values of 0 were assigned to ages due either to a large amount of controversy or because the inferences and analytical methods are not robust. This value indicates that the age is of interest and should not be dismissed, but also should not be accepted at face value. There is a high likelihood that such ages are incorrect.

For each deposit where data are available, we report our favoured age, but for some deposits, multiple ages may be available. These ages are reported in the comments section. In many cases, multiple ages support each other, lending confidence to the reported age. However, in other cases, reported ages can be significantly different. The comments section provides the alternative ages, uncertainties and sources so that the user can make their own assessment of the alternative ages.

### 3.4 Endowment

The dataset also presents estimates of endowment (i.e. pre-mining total resource estimates). These estimates include all reserves and resources in addition to production. Although grade and tonnage information were sought, it was not available in all cases, and only total metal contents are reported. Due to the manner in which some companies report resource and production data, it is not possible to assign data for projects to individual deposits. In this case the mining project was defined as a district (or part of a district) and the data are reported for the district. In some cases, it was not possible to find production data; in these cases, resource (inclusive of reserves) data are reported as the endowment, although it is noted that the endowment is under-estimated.

As part of endowment assessment, we have a column entitled “CRIRSCO-compliant” to indicate if mineral resources used to determine endowment are reported according to internationally accepted reporting codes (mostly JORC Code 2012, and also NI43-101). In this column a “Y” indicates that only code-compliant mineral resources (including measured, indicated and inferred, and including ore reserves if not included in mineral resources) were used in determining endowment.

A “P” indicates that the endowment is partially based on code-compliant mineral resources. In many of these cases, endowment is based on production data up to a certain date and mineral resources at that date.

In other cases, the endowment is based on reported code-compliant resources from which estimates of grades of unreported commodities have been estimated. The reporting of rare earth elements is a good example. Many code-compliant resource estimates only report total rare earth elements or a basket of rare earth elements (e.g., the ‘magnet basket’ – Pr, Nd, Tb and Dy). In cases where

additional geochemical data allow, the grades of all rare earth elements have been estimated, but these estimates are not code compliant.

An “N” in CRIRSCO-compliant column indicates that the endowment estimate is not code-compliant, the estimate might be based solely on production data for an exhausted resource, or the resource estimate may have predated the development of codes and not be compliant for other reasons.

Finally, an “N/A” indicates that resource and endowment data are not available.

### 3.5 Major and minor products

Based mainly on endowment, but also on the geochemical characteristics of deposit types, the commodities present or likely to be present in each deposit have been inferred. Three columns present this information: (1) Major commodities, (2) Known minor commodities, and (3) potential minor commodities. To determine the value of each commodity with endowment by using 2023 prices for the commodity presented in the 2024 edition of Mineral Commodity Summaries (United States Geological Survey, 2024). The proportion of the total value of the endowment was calculated for each commodity for which endowment data were available. Commodities that contributed over 20% of the total value were assigned as major commodities and then listed in descending order in the “Major commodities” column. Commodities contributing less than 20% of the total were listed in the “Known minor commodities” column, again in decreasing order. Appendix E contains the calculations used to determine known major and minor commodities. It must be stressed that commodities listed in these two columns can change over time and must be considered as a snapshot in time. Finally, the column entitled “Potential minor commodities” include those that are not currently produced (or reported), but based on known geochemical features of the individual deposit or deposit type may be present.

### 3.6 Host rocks

The dataset also records rocks that host the deposit. This information is presented as the rock unit name, the Geoscience Australian Stratigraphic Index number (STRATNO), a lithological description and a broad age. The lithological description can be taken either from the literature or from descriptions in Geoscience Australia’s Australian Stratigraphic Units Database ([www.asud.ga.gov.au](http://www.asud.ga.gov.au)). As some deposits have multiple host rocks, multiple (up to six) entries are allowed, with the order reflecting the relative importance of the rock unit as an ore host (most important first, least important last).

### 3.7 Associated igneous rocks

As igneous rocks are important in many mineral systems as heat, metal, sulfur and/or fluid sources, the dataset also presents information on volcanic and intrusive igneous rocks spatially or temporally associated with deposits. Similar to the recording of host rocks, for each associated igneous rock unit, the name, STRATNO, lithological descriptions and age are presented. In addition, the timing of emplacement relative to mineralisation (pre-, syn-, post- or unknown) is presented. Up to nine different entries are allowed per entity. The order reflects relative timing of emplacement (oldest first, youngest last).

## 3.8 Alteration assemblages and zonation

Information regarding regional and proximal (metamorphosed) alteration assemblages is presented as three entries for each deposit/district. The first two entries document regional and proximal alteration assemblages, whereas the third entry classifies the proximal alteration assemblages into a limited number of groups. Spatial and temporal variations (indicated by arrows) in alteration assemblages are recorded in both the regional and proximal entries. Appendix D lists abbreviations for minerals and mineral assemblages used in describing the alteration assemblages.

## 3.9 Fe-S-O minerals

The presence and relative abundance of Fe-S-O minerals (pyrite, marcasite, pyrrhotite, magnetite and hematite) within ore assemblages can provide important constraints on the initial geochemical conditions of ore formations (e.g., redox, temperature and sulfur concentrations) as well as reflecting processes that subsequently have affected the ores (e.g., metamorphism). In view of this, information has been collected on the presence and relative abundance of these minerals in hypogene ore assemblages. The dataset provides a list of these minerals as well as an indication of relative abundance. To indicate relative abundance of these minerals four operands are used: “~” indicates similar abundances of mineral A and mineral B; “>” indicates that mineral A has a greater abundance than mineral B; “>>” indicates that mineral A has a much greater abundance than mineral B; and “,” indicates the relative abundance is uncertain or unknown. Information is only provided for hypogene minerals and does not include Fe-S-O minerals formed during supergene enrichment or oxidation except in cases (e.g., paleochannel iron) in which ore formation occurred by these processes.

## 3.10 Sulfate minerals

The presence and abundance of sulfate minerals also provides information about geochemical conditions of ore formation, including the redox characteristics of mineralising fluids and the possibility of fluid mixing (in the case of barite). The dataset has compiled up to three different hypogene sulfate minerals along with their qualitative abundance in the ores (“present” indicates present at unknown abundances, “abundant” means abundant at levels of a few percent or more, “minor” indicates present at levels below a few percent, and “trace” means observed but at low levels). “Not reported” was used in cases where no evidence was found for sulfate minerals in the literature, and the literature reports were considered to be reasonably exhaustive.

## 3.11 Metamorphic grade

The metamorphic grade columns indicate the possible range of the highest metamorphic grades that has affected the ore and alteration assemblage. In most cases this is the maximum metamorphic grade to have affected the host rocks, but in a few instances, mineralising events postdated peak metamorphism. The timing of peak metamorphism relative to primary ore formation is provided when known.

## 3.12 Data sources

In this update we have divided sources of data into four categories – location, geology, age and resources/endowment. Each category has a separate column. This was done to allow readers more

ready access to specific references. Cited data sources are not intended to be exhaustive as many deposits have many tens to hundreds of papers written about them. Rather, recent summaries of the deposits have been used as important sources, supplemented by specific sources for particular information (e.g., age data). In many cases, announcements from the Australian Securities Exchange or Toronto Stock Exchange have been used and cited, particularly for mineral resources. All data sources used in the dataset are listed in a second tab of the workbook entitled “References”.

### 3.13 Comments

The comment field includes additional information that cannot be encapsulated in other fields. This information might include information on how the age of a mineralising event was established and/or supporting (or contradicting) ages or on the breakdown of endowment into production and mineral resources.

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